

Praveen

A

AI ML Engineer

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PROFILE SUMMARY

Applied AI Engineer specializing in agentic workflows and local-first LLM deployments. Track record of driving **80% infrastructure cost reductions**, optimizing RAG pipelines for **sub-100ms latency**, and shipping autonomous systems that outperform baselines by **83% in content coverage**.

CORE STACK

- **Generative AI & LLMs:** RAG, Agentic Workflows (ReAct), Prompt Engineering, Quantization (GGUF), Vector DBs (pgvector, Chroma).
- **Engineering & MLOps:** FastAPI, PostgreSQL, Kubernetes, Docker, CI/CD, MLflow, LangChain, llama.cpp.
- **Machine Learning:** PyTorch, XGBoost, Scikit-Learn, Semantic Search, Embeddings (sentence-transformers).

PROFESSIONAL EXPERIENCE

VIPS.Tech | AI/ML Engineering Intern | Jul 2025 – Dec 2025

- **Architected enterprise RAG pipelines** (LangChain, FastAPI, PostgreSQL), maintaining **99% uptime** across production environments.
- **Slashed deployment cycle times by 15%** by integrating RAGAS evaluation metrics into automated CI/CD workflows.
- **Engineered CPU-efficient quantized LLM inference** (GGUF) and a pgvector semantic retrieval pipeline, achieving **sub-100ms query latency**.

Good Shepherd Systems Services | Junior Executive – Data Operations | May 2024 – May 2025

- **Automated ETL reporting workflows** (Python, PostgreSQL) to process 500K+ monthly records, eliminating **5 hours of weekly manual operations** for leadership.
- **Boosted data quality metrics by 22%** by executing rigorous validation pipelines on 1.2M+ enterprise records.
- **Accelerated core sales and inventory query runtimes by 30%** through advanced PostgreSQL optimization.

Dhapps.ai | Data Science Intern | May 2024 – Jul 2024

- **Improved baseline prediction accuracy by 11%** by developing supervised XGBoost models on 5K+ customer records.
- **Accelerated developer onboarding by 2 days** by authoring comprehensive, end-to-end ML workflow documentation.

TECHNICAL PROJECTS

Local-First Enterprise RAG Oracle

- **Slashed annual cloud infrastructure costs by 80%** by architecting a fully offline, secure RAG system using sentence-transformers and pgvector.
- **Optimized retrieval pipelines** using quantized LLMs (GGUF via llama.cpp) for CPU-based inference, achieving **180ms average latency** on a 12GB document corpus.
- **Eliminated SaaS vendor lock-in** by migrating the core stack to open-source alternatives (LLaMA, LangChain, PostgreSQL).

Autonomous Market Research Agent (MCP-Based)

- **Engineered an autonomous ReAct AI agent** with a custom MCP server for multi-step reasoning, real-time web search, and scraping.
- **Reduced report generation time by 75%** while achieving an **83% content coverage match** and 0.81 BLEU score against human baselines.
- **Proved system scalability and fault tolerance** by simulating multi-agent deployments on Kubernetes.

Semantic Retrieval API & Offline Processing

- **Built a low-latency hybrid retrieval API** utilizing an embedding-based semantic router (MiniLM) with an LLM fallback (FLAN-T5).
- **Compressed a 14M+ word dataset (7GB) by 60%** into an optimized SQLite format with Full-Text Search, enabling sub-second queries for an Android offline dictionary.

Customer Churn Prediction Engine

- **Trained and tuned XGBoost and Random Forest models** on 7K+ records, achieving **82% predictive accuracy** for customer churn.
- **Deployed a real-time inference service** using Docker and FastAPI, enforcing strict Pydantic schema validation for reproducible business predictions.